Highlights

We will learn the following.

- States of Matter
- Change of States of Water

States of Matter

Explore More

We can observe different solids, liquids and gases around us. They exist in different states due to the difference in their properties. Let's learn more about these states of matter.

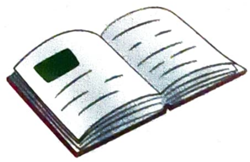
Solids

A solid is a substance whose shape cannot be changed. It also has the following features.

- It has fixed shape and size.
- It takes up space and have weight.
- It cannot flow.
- Examples: Book, table, box, cup, phone, glass

glass

table



book

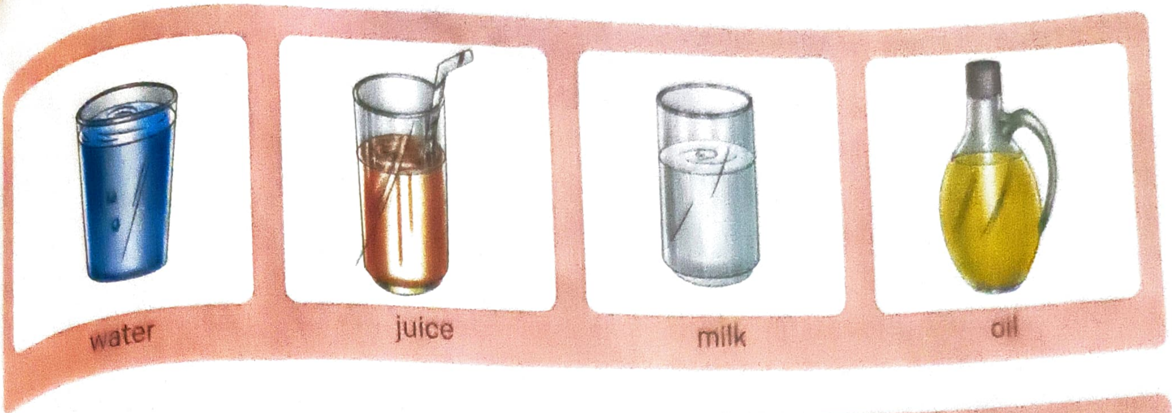
cup

Liquids

A liquid is a substance that has no shape and can take the shape of a container in which it is poured. It also has the following features.

- It has no fixed shape and size.
- It takes the shape of the container.
- It can flow easily.
- Examples: Water, milk, fruit juice, oil

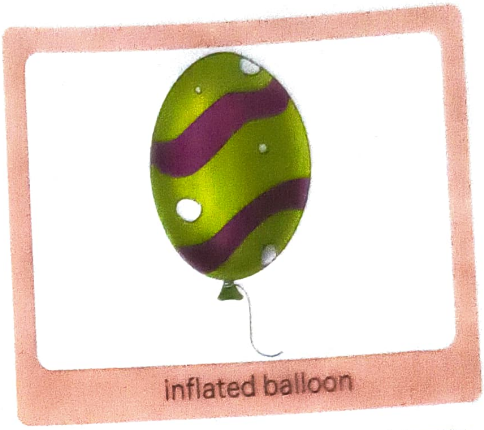
Solid: A thing that has a definite shape and size
Liquid: A thing that has no fixed shape and size, and takes the shape of the container it is poured in



Gases

A gas is a substance that has no shape and size and cannot be seen. It also has the following features.

- It has no shape or size.
- It fills all the available space.
- It is not visible.
- Examples: Oxygen gas, carbon dioxide gas


inflated balloon

Reflect

Observe the pictures of the given things carefully. Write 'S' for solids, 'L' for liquids and 'G' for gases.

L G S L S

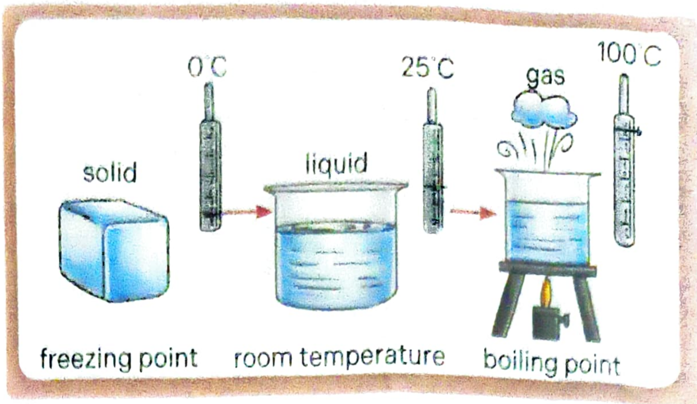
Change of States of Water

Solids, liquids and gases can interchange their states. For example, water freezes to become ice and ice melts to become water. If we heat water long enough, it turns into steam. Let's learn about the change of states of water.

Explore More

Gas: A thing that has no shape or size and fills all the available space

Different States of Water



In its solid state, water appears as ice, snow or frost. In its liquid state, it's just water. In its gaseous state, it's steam. Here are some important terms about the change of states.

Melting: It is the process in which a solid (like ice) changes to a liquid (like water) due to heating.

Freezing: It is the process in which a liquid (like water) changes to a solid (like ice) due to cooling.

Evaporation: It is the process in which a liquid (like water) changes to a gas (like steam) due to heating.

Condensation: It is the process in which a gas (like steam) changes to a liquid (like water) due to cooling.

Reflect

Write the names of the processes involved in the following.

1. When water in ice tray becomes ice Freezing
2. When drops of water form on the surface of a cold bottle Condensation
3. When the level of water in a bowl kept in the sun reduces Evaporation

Apply in Real Life

Water exists in all three forms in nature.

1. In which forms does water exist in solid state? ice.
2. Where does water exist in liquid state? Ocean
3. What is the gaseous state of water called? steam.

Skills: Critical and logical thinking, Applicative thinking

Summary Check

Different things around us exist in various forms, specifically as solids, liquids and gases. These forms can change from one form to another. For example, water can exist as a solid (ice, snow, frost), a liquid (water), and a gas (steam). This ability to change forms is a common property of many substances in our environment.

Words: gases, solid, liquids, change, steam, water

Knowledge Check

Rapid Response

A. Fill in the blanks.

1. Condensation is the process of converting steam to water.
2. A gas has no shape and size.
3. A solid has a fixed shape and ______.
4. Liquid can be poured and spilled.

B. Choose the correct answer.

1. Which of the following has a fixed shape and size?
   (a) Carbon dioxide (b) Oxygen (c) Book (d) Oil
2. Which of the following involves conversion of liquid into gas?
   (a) Evaporation (b) Condensation (c) Melting (d) Freezing
3. Which of the following can be easily poured into another container?
   (a) Eraser (b) Crayon (c) Pen (d) Oil
4. What will happen if we keep a piece of chocolate in our hand for some time?
   (a) Evaporates (b) Melts (c) Freezes (d) Condenses

C. Circle the odd one out. Give reason for your answer.

Juice Oil Book Water

In-depth Response

A. Define the following terms.

1. Melting 2. Freezing 3. Evaporation 4. Condensation

B. Answer the following questions.

1. A bottle is an example of a solid. Why?
2. How can you convert water into steam? Can steam be changed back to water? Name the process involved in both the cases.
3. How can you say that melting and freezing are opposite processes?
4. What are the features of liquids?

C. Think logically and critically to answer the following questions. (HOTS)

1. Why is water important in all its three states of matter?
2. What will happen if we take out ice cream from the freezer and keep it out?